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APPARATUS FOR PRODUCING GLASS AND METHOD FOR PRODUCING GLASS

Abstract

An apparatus for producing glass includes: a recuperator heating a heat transfer medium through heat exchange with a combustion gas emitted from a glass melting furnace; a reactor causing a reverse shift reaction to progress by heating CO.sub.2 gas and H.sub.2 gas through heat exchange with the heat transfer medium heated by the recuperator and producing at least CO gas from the CO.sub.2 gas and the H.sub.2 gas; and a burner forming a flame inside the glass melting furnace by burning a flammable gas containing the CO gas produced by the reactor and an oxidizing gas.

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Background/Summary

TECHNICAL FIELD

[0001] The present disclosure relates to an apparatus for producing glass and a method for producing glass.

BACKGROUND ART

[0002] An apparatus for producing glass includes a burner that forms a flame inside a glass melting furnace. The burner forms a flame by burning a flammable gas and an oxidizing gas. The flammable gas, for example, is natural gas. Natural gas contains CH₄ gas as its main component. The oxidizing gas, for example, is pure oxygen gas or air. The flame heats a glass raw material and molten glass that is formed by melting the glass raw material.

[0003] As described above, there are cases in which air is used as an oxidizing gas. In such cases, since N₂ gas accounting for most of air does not contribute to combustion, a large amount of air is used, and a large amount of a combustion gas is emitted from the glass melting furnace. The combustion gas is a gas remaining after a combustion reaction of a flammable gas and an oxidizing gas (including N₂ gas that does not contribute to the combustion reaction). In order to recover the heat of the combustion gas emitted in a large amount, a first regenerator and a second regenerator may be installed near the glass melting furnace.

[0004] The first regenerator and the second regenerator are used in combination with a first burner and a second burner. The apparatus for producing glass repeatedly performs formation of a flame inside the glass melting furnace using the first burner and formation of a flame inside the glass melting furnace using the second burner in an alternating manner.

[0005] The first regenerator recovers heat from the combustion gas emitted from the glass melting furnace while the second burner forms a flame inside the glass melting furnace. The first regenerator heats at least one of the flammable gas and the oxidizing gas by releasing heat recovered in advance while the first burner forms a flame inside the glass melting furnace. The first burner forms a flame inside the glass melting furnace by burning the flammable gas and the oxidizing gas at least one of which has been heated in the first regenerator.

[0006] Similarly, the second regenerator recovers heat from the combustion gas emitted from the glass melting furnace while the first burner forms a flame inside the glass melting furnace. The second regenerator heats at least one of the flammable gas and the oxidizing gas by releasing heat recovered in advance while the second burner forms a flame inside the glass melting furnace. The second burner forms a flame inside the glass melting furnace by burning the flammable gas and the oxidizing gas at least one of which has been heated in the second regenerator.

[0007] The first regenerator of Patent Document 1 dissipates heat while the first burner forms a flame inside the glass melting furnace, thereby causing an endothermic reaction to progress. As an example of the endothermic reaction, a reaction to produce CO gas and H₂ gas from CH₄ gas, H₂O gas, and CO₂ gas has been disclosed. The CO gas and the H₂ gas produced in the first regenerator are supplied to the first burner as a flammable gas.

[0008] As described above, pure oxygen gas may be used as an oxidizing gas. In this case, the oxidizing gas does not contain N₂ gas. Thus, there is a small emission amount of the combustion gas, and no regenerator is necessary, whereby the flexibility in the design of the glass melting furnace is improved. In addition, N₂ gas not contributing to combustion does not need to be heated unnecessarily, and glass can be efficiently heated using the heat of combustion. Furthermore, the

formation of NO_x gas can be suppressed.

[0009] A recuperator may be used instead of a regenerator as a means of recovering heat from the combustion gas emitted from the glass melting furnace.

[0010] A recuperator of Patent Document 2 heats air through heat exchange with a combustion gas. The air transports heat from a recuperator to a reactor. The reactor heats natural gas through heat exchange with air heated by the recuperator and causes an endothermic reaction that produces CH₄ gas from C_mH_n gas (here, m and n are integers of 2 or more) contained in the natural gas to progress.

[0011] Patent Document 3 discloses that, when the concentration of H₂O gas in the furnace atmosphere of a glass melting furnace is too high, the moisture concentration contained in molten glass becomes too high, and the quality of the glass is degraded (for example, defects are formed on the bottom surface of float glass).

CITATION LIST

Patent Document

[0012] Patent Document 1: Japanese Patent No. 3705713 [0013] Patent Document 2: Japanese Patent No. 6980795 [0014] Patent Document 3: Japanese Patent No. 6144622

SUMMARY OF INVENTION

Technical Problem

[0015] Generally, natural gas is used as a flammable gas to be burned by a burner. Natural gas contains CH₄ gas as its main component. When CH₄ gas is burned, CO₂ gas is produced and CO₂ gas is emitted. Thus, in order to reduce the amount of production of CO₂ gas and furthermore the amount of emission of CO₂ gas, H₂ gas may be considered to be used instead of CH₄ gas.

[0016] In Patent Documents 1 and 2, although a means for efficiently recovering heat of a combustion gas in a case in which CH₄ gas is used as a flammable gas has been reviewed, a means for efficiently recovering heat of a combustion gas in a case in which H₂ gas is used as a flammable gas has not been reviewed.

[0017] An aspect of the present disclosure provides a technology for reducing the specific energy consumption of molten glass in a case in which H₂ gas is used as a flammable gas.

Solution to Problem

[0018] An apparatus for producing glass according to one aspect of the present disclosure includes: a recuperator heating a heat transfer medium through heat exchange with a combustion gas emitted from a glass melting furnace; a reactor causing a reverse shift reaction to progress by heating CO₂ gas and H₂ gas through heat exchange with the heat transfer medium heated by the recuperator and producing at least CO gas from the CO₂ gas and the H₂ gas; and a burner forming a flame inside the glass melting furnace by burning a flammable gas containing the CO gas produced by the reactor and an oxidizing gas.

Advantageous Effects of Invention

[0019] According to one aspect of the present disclosure, in a case in which H₂ gas is used as a flammable gas, a reverse shift reaction of producing at least CO gas from the H₂ gas and CO₂ gas is caused to progress. The reverse shift reaction is an endothermic reaction, and thus heat of a combustion gas can be efficiently recovered. In addition, the product of an endothermic reaction (CO gas) potentially contains a larger heat content than a reactant of the endothermic reaction (the H₂ gas). Therefore, by burning CO gas, a heat of combustion per mole can be improved compared with in a case in which H₂ gas is burned. As a result, the specific energy consumption of molten glass can be reduced.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0020] FIG. 1 A diagram illustrating an apparatus for producing glass according to an embodiment.

[0021] FIG. 2 A diagram illustrating Example 1 to Example 40 of combinations of a flammable gas, a oxidizing gas, presence/absence of transformation of a flammable gas, a molar ratio x , concentration of H_{2O} gas, a reaction temperature, a reaction rate y , and a rate of increase in heat of combustion due to transformation.

[0022] FIG. 3 A diagram illustrating a first example of an apparatus for producing glass including a recuperator and a reactor.

[0023] FIG. 4 A diagram illustrating a second example of an apparatus for producing glass including a recuperator and a reactor.

[0024] FIG. 5 A diagram illustrating a third example of an apparatus for producing glass including a recuperator and a reactor.

[0025] FIG. 6 A diagram illustrating the first example of the apparatus for producing glass including a first regenerator and a second regenerator.

DESCRIPTION OF EMBODIMENTS

[0026] Hereinafter, an embodiment of the present disclosure will be described with reference to the drawings. The same reference signs will be assigned to the same components or components corresponding to each other in the drawings, and description thereof may be omitted. In the specification, “-” representing a numerical range means inclusion of numerical values using numerical values written before and after it respectively as a lower limit value and an upper limit value.

[0027] An apparatus **1** for producing glass according to an embodiment will be described with reference to FIG. **1**. The apparatus **1** for producing glass includes a glass melting furnace **10**. The glass melting furnace **10** houses a glass raw material and molten glass formed by melting the glass raw material. The molten glass is extracted from the glass melting furnace **10** and, thereafter, is formed into a desired shape and slowly cooled. In accordance with this, a glass product is acquired.

[0028] The glass raw material is prepared by mixing a plurality of types of materials. The glass raw material may contain a fining agent. The glass raw material may contain glass cullet for recycling glass. The glass raw material may be a powder material or a granulated material acquired by granulating a powder material. The glass raw material is determined in accordance with the composition of a glass product.

[0029] The glass melting furnace **10** is configured using refractories. The refractories include, for example, zirconia fused cast refractories, alumina fused cast refractories, alumina-zirconia fused cast refractories, AZS (Al—Zr—Si) fused cast refractories, dens-fired refractories, or the like. The glass melting furnace **10** may be configured using a plurality of types of refractories.

[0030] The apparatus **1** for producing glass includes a burner **20**. The burner **20** forms a flame inside the glass melting furnace **10**. The flame heats a glass raw material and molten glass. The glass raw material is fed in from above relative to the surface level of the molten glass to form a layer on at least a part of the surface level. The glass raw material is gradually fused into the molten glass. Although only one burner **20** is illustrated in FIG. **1**, commonly, a plurality of burners **20** are used.

[0031] The apparatus **1** for producing glass may use a plurality of electrodes together, which are not illustrated in the drawing, as a heat source. The plurality of electrodes apply an AC voltage to the molten glass, thereby energizing and heating the molten glass. In this case, the molten glass produces heat. The apparatus **1** for producing glass may additionally use an electric heater, which is not illustrated in the drawing, as a heat source. The electric heater produces heat itself. The apparatus **1** for producing glass includes at least the burner **20** as a heat source.

[0032] The burner **20** forms a flame by burning a flammable gas and a oxidizing gas. In this embodiment, H_2 gas (in more detail, a gas acquired by transforming to H_2 gas as will be described below) is used as at least a part of the flammable gas. By using the H_2 gas, compared to a case in which CH_4 gas is used, the amount of production of CO_2 gas and, furthermore, the amount of emission of CO_2 gas can be reduced.

[0033] H_2 gas may be used as at least a part of the flammable gas, and natural gas may be used in combination therewith. Since natural gas contains CH_4 gas as its main component, the less natural gas is used, the better. From the point of view of reducing the amount of production of CO_2 gas, the amount of use of natural gas is preferably between 0 volume % to 50 volume %, more preferably

between 0 volume % to 25 volume %, and even more preferably 0 volume %.

[0034] When H.sub.2 gas is burned instead of CH.sub.4 gas, H.sub.2O gas is produced through a reaction of H.sub.2 gas and O.sub.2 gas and no CO.sub.2 gas is produced. However, the concentration of the H.sub.2O gas in the furnace atmosphere rises by an amount corresponding to no production of CO.sub.2 gas. This is significant in a case in which pure oxygen gas is used as an oxidizing gas. In a case in which pure oxygen gas is used, the concentration of H.sub.2O gas in the furnace atmosphere theoretically becomes 100 volume %.

[0035] The glass raw material may emit a gas at the time of being melted. For example, in a case in which the glass raw material contains a carbonate such as calcium carbonate or magnesium carbonate, the glass raw material emits CO.sub.2 gas at the time of melting. Thus, the concentration of H.sub.2O gas in the furnace atmosphere actually becomes less than 100 volume %.

[0036] In a case in which the concentration of H.sub.2O gas in the furnace atmosphere is too high, NaOH gas, which originates from Na contained in the molten glass, is readily produced, and a furnace material such as a brick readily corrodes. Alternatively, in a case in which the concentration of H.sub.2O gas in the furnace atmosphere is too high, the concentration of moisture contained in the molten glass becomes too high, and the quality of glass may be degraded.

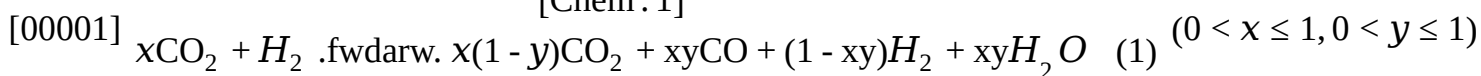
[0037] When air is used instead of pure oxygen gas as an oxidizing gas, the concentration of H.sub.2O gas in the furnace atmosphere can be reduced. In this case, since N.sub.2 gas, which accounts for most of air, does not contribute to combustion, a large amount of air is used, and a large amount of a combustion gas is emitted from the glass melting furnace. The combustion gas is a gas remaining after a combustion reaction between a flammable gas and an oxidizing gas (also including N.sub.2 gas that does not contribute to combustion). In order to recover the heat of a combustion gas emitted in a large amount, a regenerator is disposed near the glass melting furnace **10**.

[0038] On the other hand, when pure oxygen gas is used as an oxidizing gas, the oxidizing gas does not contain N.sub.2 gas. Thus, the amount of emission of the combustion gas is small, and no regenerator becomes necessary, and thus the flexibility in the design of the glass melting furnace **10** becomes high. In addition, N.sub.2 gas, which does not contribute to combustion, does not need to be heated unnecessarily, and the glass can be efficiently heated with heat of combustion. Furthermore, the generation of NO_x gas can be suppressed.

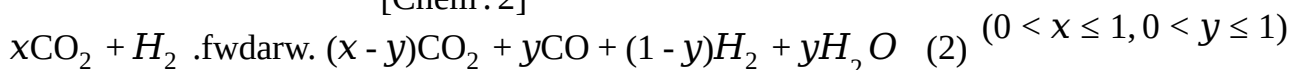
[0039] However, when pure oxygen gas is used as an oxidizing gas, compared to a case in which air is used, the concentration of H.sub.2O gas in the furnace atmosphere becomes high. Also in a case in which oxygen-enriched air is used instead of pure oxygen gas as an oxidizing gas, compared to a case in which air is used, the concentration of H.sub.2O gas becomes high. The oxygen-enriched air is a mixed gas of pure oxygen gas and air and is a gas having a higher concentration of oxygen gas than the air.

[0040] The apparatus **1** for producing glass according to this embodiment includes a gas transforming unit **30** that transforms H.sub.2 gas for reducing the concentration of H.sub.2O gas in the furnace atmosphere. One or more (one in FIG. **1**) gas transforming units **30** are disposed. The gas transforming unit **30** produces at least CO gas and H.sub.2O gas from CO.sub.2 gas and H.sub.2 gas through a reverse shift reaction (see Equation (1) and Equation (2) represented below). CO.sub.2 gas and H.sub.2 gas are reactants and CO gas and H.sub.2O gas are products. The products may further contain at least one of H.sub.2 gas and CO.sub.2 gas.

[Chem. 1]



[Chem. 2]



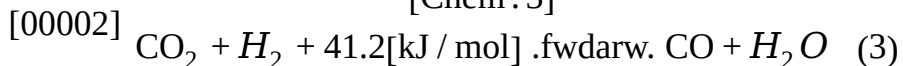
[0041] In Equation (1) and Equation (2), x represents the molar ratio of CO.sub.2 gas to H.sub.2 gas supplied to the gas transforming unit **30**. The molar ratio x is greater than 0.00. In general, the molar ratio is equal to a volume ratio. y represents the reaction rate of a reverse shift reaction. The reaction rate y being 0.00 means that the reverse shift reaction has not progressed at all. The reaction rate y

being 1.00 means that the reverse shift reaction has been completed. The reaction rate y represents a ratio of the production amount of CO gas (in moles) to a maximum production amount of CO gas (in moles). The maximum production amount of CO gas is equal to a minimum value of the initial amount of CO.sub.2 gas and the initial amount of H.sub.2 gas.

[0042] The gas transforming unit **30** may cause a reverse shift reaction to progress and may cause a reaction different from the reverse shift reaction to simultaneously progress. Thus, CO.sub.2 gas and H.sub.2 gas may be supplied to the gas transforming unit **30** as reactants, and any other gas may be further supplied thereto.

[0043] The reverse shift reaction is represented using the following Equation (3).

[Chem . 3]



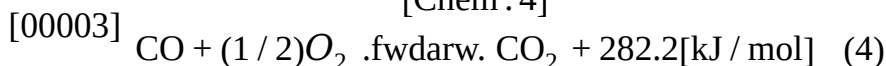
[0044] From Equation (3), it can be understood that the reverse shift reaction is an endothermic reaction. In an endothermic reaction, the higher the temperature of the reaction field, the more the reaction progresses. A product of the endothermic reaction (CO gas) potentially contains a greater heat content than the reactant of the endothermic reaction (H.sub.2 gas). Thus, when the product of the endothermic reaction (CO gas) is burned, a greater heat of combustion than that of a heat of combustion of the reactant of the endothermic reaction (H.sub.2 gas) can be acquired.

[0045] The gas transforming unit **30** supplies products such as CO gas and the like to the burner **20**. The supplied gases include, for example, CO gas and H.sub.2O gas (see Equation (1) and Equation (2)). The supplied gases may additionally contain at least one of H.sub.2 gas and CO.sub.2 gas.

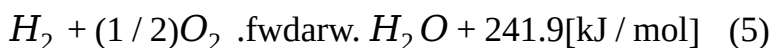
[0046] The burner **20** forms a flame inside the glass melting furnace **10** by burning a flammable gas including the CO gas produced by the gas transforming unit **30** and an oxidizing gas. By burning CO gas, the heat of combustion per mole can be improved compared with the heat of combustion of H.sub.2 gas.

[0047] The combustion reaction of CO gas is represented using the following Equation (4), and the combustion reaction of H.sub.2 gas is represented using the following Equation (5). Here, the heat of combustion of H.sub.2 gas is expressed using a lower heating value.

[Chem . 4]



[Chem . 5]

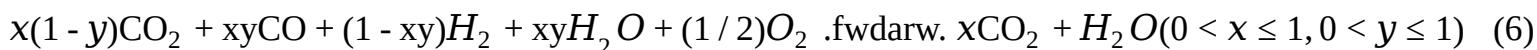


[0048] From Equation (4) and Equation (5), it can be understood that, by burning CO gas, the heat of combustion per mole can be improved compared with in a case in which H.sub.2 gas is burned.

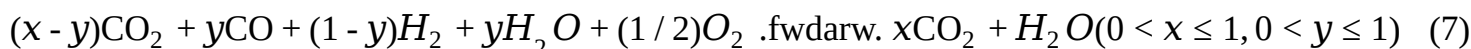
[0049] A combustion reaction of a flammable gas and an oxidizing gas that is performed using the burner **20** is represented, for example, using the following Equation (6) and the following Equation (7).

[00004]

[Chem . 6]



[Chem . 7]



[0050] Equation (6) represents a combustion reaction of the product acquired in Equation (1). On the other hand, Equation (7) represents a combustion reaction of the product acquired in Equation (2). In Equation (6) and Equation (7), x represents the molar ratio of CO.sub.2 gas to H.sub.2 gas supplied to the gas transforming unit **30**. The molar ratio x is greater than 0.00.

[0051] The combustion reaction of a flammable gas and an oxidizing gas may include at least the combustion reactions represented in Equation (6) and Equation (7) and may further include yet another combustion reaction.

[0052] The burner **20**, as described above, burns a flammable gas including CO gas produced by the gas transforming unit **30** and an oxidizing gas. Thus, the combustion gases include not only H.sub.2O

gas but also CO.sub.2 gas. This is apparent also from Equation (6) and Equation (7). The CO.sub.2 gas can dilute the H.sub.2O gas and reduce the concentration of H.sub.2O gas in the furnace atmosphere. [0053] FIG. 2 illustrates Example 1 to Example 40 of combinations of a flammable gas, an oxidizing gas, presence/absence of transformation of the flammable gas, a molar ratio x, concentration of H.sub.2O gas, a reaction temperature, a reaction rate y, and the rate of increase R in heat of combustion due to transformation. In FIG. 2, “the concentration of H.sub.2O gas” is the concentration of H.sub.2O gas in the combustion gas. The “reaction temperature” is a reaction temperature in the gas transforming unit 30. The “reaction rate y” is a value acquired when the reverse shift reaction reaches equilibrium and is a value without taking a reaction speed into account. In a case in which the molar ratio x is constant, the reaction rate y changes in accordance with the reaction temperature, and the higher the reaction temperature, the higher the reaction rate y. “The rate of increase R in heat of combustion due to transformation” represents the ratio (%) of the heat of combustion of the flammable gas after transformation to the heat of combustion of the flammable gas before transformation. [0054] In a case in which the molar ratio x is 1.00 or less, the rate of increase R is obtained using the following Equation (8).

[Math. 1]

$$[00005] \quad R = (241.9 + 41.2xy) / 241.9 \times 100 \quad (8)$$

[0055] Equation (8) represents the ratio of the heat of combustion of a flammable gas after transformation (xy moles of CO gas and (1-xy) moles of H.sub.2 gas) to the heat of combustion of a flammable gas before transformation (1 mole of H.sub.2 gas) (see Equation (4), Equation (5), and Equation (6)). As is apparent from Equation (8), in a case in which the molar ratio x is constant, the higher the reaction rate y, the larger the rate of increase R.

[0056] In a case in which the molar ratio x is greater than 1.00, the rate of increase R is obtained using the following Equation (9).

[Math. 2]

$$[00006] \quad R = (241.9 + 41.2y) / 241.9 \times 100 \quad (9)$$

[0057] Equation (9) represents the ratio (%) of the heat of combustion of a flammable gas after transformation (y moles of CO gas and (1-y) moles of H.sub.2 gas) to the heat of combustion of the flammable gas before transformation (1 mole of H.sub.2 gas) (see Equation (4), Equation (5), and Equation (7)). As is apparent from Equation (9), regardless of the molar ratio x, the higher the reaction rate y, the larger the rate of increase R.

[0058] As illustrated in FIG. 2, when H.sub.2 gas is used instead of CH.sub.4 gas as the flammable gas, the concentration of H.sub.2O gas in the furnace atmosphere becomes higher in accordance with no production of CO.sub.2 gas. This becomes remarkable in a case in which a pure oxygen gas is used as an oxidizing gas. In a case in which the pure oxygen gas is used, the concentration of the H.sub.2O gas in the combustion gas is 100 volume %.

[0059] The glass raw material may emit a gas when it is melted. Thus, the concentration of the H.sub.2O gas in the furnace atmosphere is actually less than 100 volume %. Although the concentration of H.sub.2O gas in the furnace atmosphere depends also on the type of glass raw material, it is determined mainly in accordance with types of flammable gas and oxidizing gas. Therefore, in this embodiment, the concentration of the H.sub.2O gas in the combustion gas is employed as a parameter to be managed.

[0060] The concentration of the H.sub.2O gas in the combustion gas is preferably less than 67 volume %. Conventionally, molten glass has been produced by burning CH.sub.4 gas and O.sub.2 gas (a pure oxygen gas), and, it has been confirmed that there are no problems regarding the quality of a glass product and corrosion of a furnace material when the concentration of H.sub.2O gas in the combustion gas is 67 volume % or less.

[0061] It is preferable that the molar ratio x should be 0.50 or more such that the concentration of H.sub.2O gas in the combustion gas is 67 volume % or less (see FIG. 2). It is more preferable that the concentration of the H.sub.2O gas in the combustion gas should be 65 volume % or less. It is more preferable that the molar ratio x should be 0.55 or more such that the concentration of H.sub.2O gas in

the combustion gas is 65 volume % or less. Although a lower limit of the concentration of the H.sub.2O gas in the combustion gas is not particularly limited, from the point of view of feasibility, it is preferable that the lower limit is 20 volume % or more. It is preferable that the molar ratio x should be 4.00 or less such that the concentration of the H.sub.2O gas in the combustion gas becomes 20 volume % or more.

[0062] The molar ratio x should be 1.00 or less. If the molar ratio x is 1.00 or less, the combustion reaction is represented by Equation (6). On the other hand, if the molar ratio x exceeds 1.00, the combustion reaction is represented by Equation (7). As is apparent from Equation (7), if the molar ratio x exceeds 1.00, there are $(x-y)$ moles of the CO.sub.2 gas not contributing to combustion. This CO.sub.2 gas not only does not produce heat of combustion but also wastes heat of combustion. If the molar ratio x is less than 1.00, the amount of the CO.sub.2 gas that wastes heat of combustion can be reduced, and glass can be heated efficiently with the heat of combustion.

[0063] In supplying CO.sub.2 gas to the gas transforming unit **30**, although a dedicated supply source such as a gas cylinder that is not illustrated in the drawing may be used, in this embodiment, a circulation line **50** is used. The circulation line **50** supplies at least a part of the CO.sub.2 gas contained in the combustion gas from the glass melting furnace **10** to the gas transforming unit **30**. At least a part of the CO.sub.2 gas emitted from the glass melting furnace **10** can be used as a reactant of a reverse shift reaction. The CO.sub.2 gas can be circulated, and the amount of emission of the CO.sub.2 gas can be reduced.

[0064] It is preferable that a dehydration unit **51** is disposed in the middle of the circulation line **50**. The dehydration unit **51** removes H.sub.2O gas from the combustion gas, thereby separating the H.sub.2O gas and the CO.sub.2 gas from each other. At least a part of the separated CO.sub.2 gas can be used as a reactant in a reverse shift reaction. The supply of the H.sub.2O gas to the gas transforming unit **30** can be limited, and the concentration of the H.sub.2O gas in the furnace atmosphere of the glass melting furnace **10** can be reduced. Although the dehydration unit **51** may be configured to remove at least a part of the H.sub.2O gas, in this embodiment, it substantially removes all of the H.sub.2O gas.

[0065] It is preferable that the apparatus **1** for producing glass should recover heat from the combustion gas burned by the burner **20** and cause the reverse shift reaction to progress using the recovered heat. By recovering heat from the burned combustion gas, the waste of energy can be reduced. As a heat recovering means, a recuperator or a regenerator to be described below is used. Hereinafter, the heat recovering means and the like will be described.

[0066] An apparatus **1** for producing glass according to a first example that includes a recuperator **60** and a reactor **70** will be described with reference to FIG. **3**. The apparatus **1** for producing glass includes a glass melting furnace **10**, a burner **20**, a recuperator **60**, and a reactor **70**. The reactor **70** is an example of the gas transforming unit **30** and, for example, causes a reverse shift reaction to progress. Hereinafter, differences from the apparatus **1** for producing glass illustrated in FIG. **1** will be mainly described.

[0067] The recuperator **60** heats a heat transfer medium through heat exchange with the combustion gas emitted from the glass melting furnace **10**. Although the heat transfer medium may be either a gas or a liquid, the temperature of the combustion gas is high, and thus it is preferable that the heat transfer medium should be a gas from the point of view of prevention of boiling. The heat transfer medium is preferably the air. The recuperator **60** transfers heat between the combustion gas and the heat transfer medium (for example, air) without mixing them. Although the recuperator **60** is disposed outside the glass melting furnace **10** in FIG. **3**, it may be disposed inside the glass melting furnace **10**. One or more (one in FIG. **3**) recuperators **60** are disposed. The recuperator **60** is thermally connected to the reactor **70** through a heat transport line **80**. The heat transport line **80** transports the heat transfer medium from the recuperator **60** to reactor **70**.

[0068] The reactor **70** causes a reverse shift reaction to progress by heating CO.sub.2 gas and H.sub.2 gas through heat exchange with the heat transfer medium heated by the recuperator **60** and produces at least CO gas from CO.sub.2 gas and H.sub.2 gas. The CO.sub.2 gas and the H.sub.2 gas are reactants, and the CO gas and the H.sub.2O gas are products (see Equation (3)). The products may further

contain H.sub.2 gas or CO.sub.2 gas (see Equation (1) and Equation (2)). The reactor **70** does not mix the reactants and the products of the reverse shift reaction and the heat transfer medium (for example, the air) and transfers heat between them.

[0069] Since the reverse shift reaction is an endothermic reaction, heat can be efficiently recovered from the heat transfer medium, and the heat recovered from the combustion gas can be efficiently used. In addition, the product of the endothermic reaction (CO gas) potentially contains a larger heat content than the reactant of the endothermic reaction (H.sub.2 gas). Hence, combustion of CO gas provides more heat of combustion per mole than combustion of H.sub.2 gas (see equations (4) and (5)). As a result, the specific energy consumption of molten glass can be reduced. The specific energy consumption is an amount of the consumption of energy input from outside the system for producing one ton of molten glass.

[0070] A reaction occurring in reactor **70** may be an endothermic reaction as a whole and may include a reaction other than the reverse shift reaction. Thus, in the reactor **70**, reactants and products are not particularly limited. The product (that is, the flammable gas) produced in the reactor **70** is supplied to the burner **20** through a first supply line **21**. On the other hand, the oxidizing gas is supplied to the burner **20** through a second supply line **22**.

[0071] It is preferable that the apparatus **1** for producing glass should include a circulation line **50**. The circulation line **50** supplies at least a part of CO.sub.2 gas contained in the combustion gas burned by the burner **20** from the glass melting furnace **10** to the reactor **70**. At least a part of the CO.sub.2 gas emitted from the glass melting furnace **10** can be used as a reactant of the reverse shift reaction. The CO.sub.2 gas can be circulated, and the amount of emission of the CO.sub.2 gas can be reduced.

[0072] Apparatus **1** for producing glass preferably comprises a dehydration unit **51** in the middle of circulation line **50**. The dehydration unit **51** separates the H.sub.2O gas from the CO.sub.2 gas by removing the H.sub.2O gas from the combustion gas. The circulation of H.sub.2O gas can be restricted while permitting the circulation of CO.sub.2 gas. H.sub.2O gas not only does not generate heat of combustion but also wastes heat of combustion. If the circulation of H.sub.2O gas can be restricted, the non-flammable H.sub.2O gas can be avoided, and the heat of combustion can be used to heat glass efficiently. If the circulation of H.sub.2O gas can be restricted, the concentration of H.sub.2O gas in the furnace atmosphere can be reduced. The dehydration unit **51** may remove at least a portion of the H.sub.2O gas, but in the present embodiment substantially all of the H.sub.2O gas is removed.

[0073] It is more preferable that a desulfurization unit **52** should be disposed in the middle of the circulation line **50**. The desulfurization unit **52** removes a sulfur-containing gas from the combustion gas. Sulfur contained in the sulfur-containing gas is originated from a glass raw material and, for example, is originated from a fining agent. By disposing the desulfurization unit **52** in the middle of the circulation line **50**, the supply of the sulfur-containing gas to the reactor **70** can be restricted. In accordance with this, the reactor **70** can be maintained to be in a clean state.

[0074] Unlike the regenerator to be described below, the recuperator **60** and the reactor **70** recover heat of the combustion gas using a heat transfer medium, and thus a combustion gas can be restricted from directly flowing to the reactor **70**, and the reactor **70** can be maintained to be in a clean state. Thus, in a case in which the reactor **70** has a metal catalyst, the degradation of that metal catalyst can be suppressed.

[0075] The reactor **70** may have a metal catalyst. The metal catalyst promotes a reverse shift reaction. The metal catalyst can cause the reaction to reach equilibrium in a short time by increasing the reaction speed of the reverse shift reaction. The metal catalyst can also reduce the volume of reactor **70** by increasing the reaction speed of the reverse shift reaction. Copper (Cu), nickel (Ni), or the like is used as the metal catalyst. The metal catalyst may be omitted if the reaction speed is sufficiently fast.

[0076] An apparatus **1** for producing glass according to a second example that includes a recuperator **60** and a reactor **70** will be described with reference to FIG. **4**. Hereinafter, differences from the apparatus **1** for producing glass illustrated in FIG. **3** will be mainly described. As illustrated in FIG. **4**, it is preferable that the apparatus **1** for producing glass should include a heat exchanger **23**.

[0077] The heat exchanger **23** is disposed in the middle of the second supply line **22** and preheats a oxidizing gas (for example, a pure oxygen gas) through heat exchange with a heat transfer medium

heated by the recuperator **60**. The heat recovered from a combustion gas can be effectively used, and thus the specific energy consumption of molten glass can be reduced. The heat exchanger **23** transfers heat between the oxidizing gas and the heat transfer medium (for example, the air) without mixing them.

[0078] The heat transport line **80** transports the heat transfer medium from the recuperator **60** to the heat exchanger **23**. Preferably, the heat transport line **80** transports the heat transfer medium from the recuperator **60** to the reactor **70** and then transports the heat transfer medium from the reactor **70** to the heat exchanger **23**. Since the reactants of the reverse shift reaction can be heated before heating the oxidizing gas, heat that is necessary for causing the reverse shift reaction to progress can be easily secured.

[0079] An apparatus **1** for producing glass according to a third example that includes a recuperator **60** and a reactor **70** will be described with reference to FIG. 5. Hereinafter, differences from the apparatus **1** for producing glass illustrated in FIG. 3 will be mainly described. As illustrated in FIG. 5, it is preferable that the apparatus **1** for producing glass should include a second heat exchanger **24** and a second dehydration unit **25**.

[0080] The second heat exchanger **24** transfers heat from a product of the reactor **70** to the oxidizing gas at a first place of the first supply line **21** that overlaps the second supply line **22**. The second heat exchanger **24** transfers heat between the product of the reactor **70** and the oxidizing gas without mixing them. The second dehydration unit **25** removes H.sub.2O gas from the product of the reactor **70** at a second place disposed on the downstream of the first place of the first supply line **21**. Although the second dehydration unit **25** may remove at least a part of the H.sub.2O gas, in this example, it substantially removes the whole H.sub.2O gas.

[0081] The second dehydration unit **25** removes H.sub.2O gas from the product of the reactor **70**, for example, by cooling the product of the reactor **70**. The H.sub.2O gas not only does not produce heat of combustion but wastes the heat of combustion. The second dehydration unit **25** restricts the supply of the H.sub.2O gas to the burner **20**. The H.sub.2O gas not contributing to combustion does not need to be heated unnecessarily, and glass can be efficiently heated with the heat of combustion. In addition, the concentration of the H.sub.2O gas in the furnace atmosphere can be reduced.

[0082] Before the second dehydration unit **25** cools the product of the reactor **70**, the second heat exchanger **24** transfers heat from the product of the reactor **70** to the oxidizing gas. Therefore, the heat recovered from the combustion gas can be efficiently used, and the specific energy consumption of molten glass can be reduced.

[0083] An apparatus **1** for producing glass according to the first example that includes a first regenerator **90A** and a second regenerator **90B** will be described with reference to FIG. 6. The apparatus **1** for producing glass includes a glass melting furnace **10**, a first burner **20A**, a second burner **20B**, a first regenerator **90A**, and a second regenerator **90B**. The first regenerator **90A** and the second regenerator **90B** are examples of the gas transforming unit **30** and, for example, cause a reverse shift reaction to progress. Hereinafter, differences from the apparatus **1** for producing glass illustrated in FIG. 1 will be mainly described.

[0084] The first regenerator **90A** and the second regenerator **90B** are used in combination with the first burner **20A** and the second burner **20B**. Although one first regenerator **90A** is illustrated in FIG. 6, a plurality of first regenerators **90A** may be provided. Similarly, the number of second regenerators **90B**, the number of first burners **20A**, and the number of second burners **20B** may be one or more.

[0085] The apparatus **1** for producing glass alternately repeats formation of a flame inside the glass melting furnace **10** using the first burner **20A** and formation of a flame inside the glass melting furnace **10** using the second burner **20B**. FIG. 6 is a diagram illustrating a state in which the first burner **20A** forms a flame inside the glass melting furnace **10**.

[0086] While the second burner **20B** forms a flame inside the glass melting furnace **10**, the first regenerator **90A** recovers heat from the combustion gas emitted from the glass melting furnace **10**. While the first burner **20A** forms a flame inside the glass melting furnace **10**, the first regenerator **90A** emits heat that has been recovered in advance and causes a reverse shift reaction producing at least CO gas from CO.sub.2 gas and H.sub.2 gas to progress. The first burner **20A** forms a flame inside the glass

melting furnace **10** by burning a flammable gas including the CO gas produced by the first regenerator **90A** and an oxidizing gas.

[0087] Similarly, while the first burner **20A** forms a flame inside the glass melting furnace **10**, the second regenerator **90B** recovers heat from the combustion gas emitted from the glass melting furnace **10**. While the second burner **20B** forms a flame inside the glass melting furnace **10**, the second regenerator **90B** emits heat that has been recovered in advance and causes a reverse shift reaction producing at least CO gas from CO.sub.2 gas and H.sub.2 gas to progress. The second burner **20B** forms a flame inside the glass melting furnace **10** by burning a flammable gas including the CO gas produced by the second regenerator **90B** and an oxidizing gas.

[0088] The first regenerator **90A** and the second regenerator **90B** are different from the recuperator **60** and the reactor **70** in that the combustion gas flows directly into them. The heat of the combustion gas is accumulated in furnace beds and the like of the first regenerator **90A** and the second regenerator **90B**.

[0089] The first regenerator **90A** and the second regenerator **90B** cause a reverse shift reaction to progress by heating CO.sub.2 gas and H.sub.2 gas to produce at least CO gas from the CO.sub.2 gas and the H.sub.2 gas. The CO.sub.2 gas and the H.sub.2 gas are reactants, and the CO gas and the H.sub.2O gas are products (see Equation (3)). The products may further contain H.sub.2 gas or CO.sub.2 gas (see Equations (1) and (2)).

[0090] Since the reverse shift reaction is an endothermic reaction, heat can be efficiently recovered from the combustion gas, and the heat of the combustion gas can be efficiently used. In addition, the product of the endothermic reaction (the CO gas) potentially contains a larger heat content than the reactant of the endothermic reaction (the H.sub.2 gas). Thus, by burning CO gas, heat of combustion per mole can be improved more than that of combustion of H.sub.2 gas (see Equations (4) and (5)). As a result, the specific energy consumption of molten glass can be reduced.

[0091] Reactions occurring in the first regenerator **90A** and the second regenerator **90B** may be an endothermic reaction as a whole and may include a reaction other than the reverse shift reaction. Thus, the reactants and the products in the first regenerator **90A** and the second regenerator **90B** are not particularly limited.

[0092] It is preferable that the apparatus **1** for producing glass should include a circulation line **50**. While the first burner **20A** forms a flame inside the glass melting furnace **10**, the circulation line **50** supplies at least a part of the CO.sub.2 gas contained in the combustion gas emitted from the glass melting furnace **10** from the second regenerator **90B** to the first regenerator **90A**. At least a part of the CO.sub.2 gas emitted from the glass melting furnace **10** can be used as a reactant of the reverse shift reaction. The CO.sub.2 gas can be circulated, and thus the amount of emission of the CO.sub.2 gas can be reduced.

[0093] While the second burner **20B** forms a flame inside the glass melting furnace **10**, the circulation line **50** may supply at least a part of the CO.sub.2 gas contained in the combustion gas emitted from the glass melting furnace **10** from the first regenerator **90A** to the second regenerator **90B**. Although the circulation line **50** circulating the CO.sub.2 gas while the first burner **20A** forms a flame inside the glass melting furnace **10** and the circulation line **50** circulating the CO.sub.2 gas while the second burner **20B** forms a flame inside the glass melting furnace **10** are the same in this example, they may be different from each other and may be separately disposed.

[0094] Apparatus **1** for producing glass preferably comprises a dehydration unit **51** in the middle of circulation line **50**. The dehydration unit **51** separates the H.sub.2O gas from the CO.sub.2 gas by removing the H.sub.2O gas from the combustion gas. The circulation of H.sub.2O gas can be restricted while permitting the circulation of CO.sub.2 gas. H.sub.2O gas not only does not generate heat of combustion but also wastes heat of combustion. If the circulation of H.sub.2O gas can be restricted, the non-flammable H.sub.2O gas can be avoided, and the heat of combustion can be used to heat glass efficiently. If the circulation of H.sub.2O gas can be restricted, the concentration of H.sub.2O gas in the furnace atmosphere can be reduced. The dehydration unit **51** may remove at least a portion of the H.sub.2O gas, but in the present embodiment substantially all of the H.sub.2O gas is removed.

[0095] It is more preferable that a desulfurization unit **52** should be disposed in the middle of the

circulation line 50. The desulfurization unit 52 removes a sulfur-containing gas from the combustion gas. Sulfur contained in the sulfur-containing gas is originated from a glass raw material and, for example, is originated from a fining agent. By disposing the desulfurization unit 52 in the middle of the circulation line 50, the supply of the sulfur-containing gas to the first regenerator 90A or the second regenerator 90B can be restricted.

[0096] In relation with the embodiments and the like described above, the following supplementary notes are disclosed.

[Supplementary Note 1]

[0097] An apparatus for producing glass, the apparatus including: a recuperator heating a heat transfer medium through heat exchange with a combustion gas emitted from a glass melting furnace; a reactor causing a reverse shift reaction to progress by heating CO.sub.2 gas and H.sub.2 gas through heat exchange with the heat transfer medium heated by the recuperator and producing at least CO gas from the CO.sub.2 gas and the H.sub.2 gas; and a burner forming a flame inside the glass melting furnace by burning a flammable gas containing the CO gas produced by the reactor and an oxidizing gas.

[Supplementary Note 2]

[0098] The apparatus for producing glass described in Supplementary Note 1, in which the oxidizing gas is a pure oxygen gas or oxygen-enriched air.

[Supplementary Note 3]

[0099] The apparatus for producing glass described in Supplementary Note 1 or 2, further including a circulation line supplying at least a part of the CO.sub.2 gas contained in the combustion gas burned by the burner from the glass melting furnace to the reactor.

[Supplementary Note 4]

[0100] The apparatus for producing glass described in Supplementary Note 3, further including a dehydration unit, which is disposed in the middle of the circulation line, removing H.sub.2O gas from the combustion gas.

[Supplementary Note 5]

[0101] The apparatus for producing glass described in any one of Supplementary Notes 1 to 4, further including a heat exchanger preheating the oxidizing gas through heat exchange with the heat transfer medium heated by the recuperator.

[Supplementary Note 6]

[0102] The apparatus for producing glass described in any one of Supplementary Notes 1 to 5, further including: a first supply line supplying a product of the reactor to the burner; a second supply line supplying the oxidizing gas to the burner; a second heat exchanger transferring heat from the product of the reactor to the oxidizing gas at a first place of the first supply line that overlaps the second supply line; and a second dehydration unit removing H.sub.2O gas from the product of the reactor at a second place disposed on the downstream of the first place of the first supply line.

[Supplementary Note 7]

[0103] The apparatus for producing glass described in any one of Supplementary Notes 1 to 6, in which the reactor has a metal catalyst promoting the reverse shift reaction.

[Supplementary Note 8]

[0104] An apparatus for producing glass that alternately repeats formation of a flame inside a glass melting furnace using a first burner and formation of a flame inside the glass melting furnace using a second burner, the apparatus for producing glass including: a first regenerator causing a reverse shift reaction of producing at least CO gas from CO.sub.2 gas and H.sub.2 gas to progress by recovering heat from a combustion gas emitted from the glass melting furnace while the second burner forms a flame inside the glass melting furnace and by discharging heat while the first burner forms a flame inside the glass melting furnace; the first burner forming a flame inside the glass melting furnace by burning a flammable gas containing the CO gas produced in the first regenerator and an oxidizing gas; a second regenerator causing a reverse shift reaction of producing at least CO gas from CO.sub.2 gas and H.sub.2 gas to progress by recovering heat from a combustion gas emitted from the glass melting furnace while the first burner forms a flame inside the glass melting furnace and by discharging heat while the second burner forms a flame inside the glass melting furnace; and the second burner forming

a flame inside the glass melting furnace by burning a flammable gas containing the CO gas produced in the second regenerator and an oxidizing gas.

[Supplementary Note 9]

[0105] The apparatus for producing glass described in Supplementary Note 8, in which the oxidizing gas is a pure oxygen gas or oxygen-enriched air.

[Supplementary Note 10]

[0106] The apparatus for producing glass described in Supplementary Note 8 or 9, further including a circulation line supplying at least a part of the CO.sub.2 gas contained in the combustion gas emitted from the glass melting furnace from the second regenerator to the first regenerator while the first burner forms a flame inside the glass melting furnace.

[Supplementary Note 11]

[0107] The apparatus for producing glass described in Supplementary Note 10, further including a dehydration unit, which is disposed in the middle of the circulation line, removing H.sub.2O gas from the combustion gas.

[Supplementary Note 12]

[0108] A method for producing glass including producing molten glass using the apparatus for producing glass described in any one of Supplementary Notes 1 to 11.

[0109] Although the apparatuses for producing glass and the methods for producing glass according to the present disclosure have been described as above, the present disclosure is not limited to the embodiment and the like described above. Various changes, modifications, substitutions, additions, deletions, and combinations can be made within the scope of the claims. It is apparent that those belong to the technical scope of the present disclosure.

[0110] Priority is claimed on Japanese Patent Application No. 2022-188269, filed Nov. 25, 2022, the content of which is incorporated herein by reference.

REFERENCE SIGNS LIST

[0111] **1** Apparatus for producing glass [0112] **10** Glass melting furnace [0113] **20** Burner [0114] **30** Gas transforming unit

Claims

1. An apparatus for producing glass, the apparatus comprising: a recuperator heating a heat transfer medium through heat exchange with a combustion gas emitted from a glass melting furnace; a reactor causing a reverse shift reaction to progress by heating CO.sub.2 gas and H.sub.2 gas through heat exchange with the heat transfer medium heated by the recuperator and producing at least CO gas from the CO.sub.2 gas and the H.sub.2 gas; and a burner forming a flame inside the glass melting furnace by burning a flammable gas containing the CO gas produced by the reactor and an oxidizing gas.
2. The apparatus for producing glass according to claim 1, wherein the oxidizing gas is pure oxygen gas or oxygen-enriched air.
3. The apparatus for producing glass according to claim 1, further comprising a circulation line supplying at least a part of the CO.sub.2 gas contained in the combustion gas burned by the burner from the glass melting furnace to the reactor.
4. The apparatus for producing glass according to claim 3, further comprising a dehydration unit, which is disposed in the middle of the circulation line, removing H.sub.2O gas from the combustion gas.
5. The apparatus for producing glass according to claim 1, further comprising a heat exchanger preheating the oxidizing gas through heat exchange with the heat transfer medium heated by the recuperator.
6. The apparatus for producing glass according to claim 1, further comprising: a first supply line supplying a product of the reactor to the burner; a second supply line supplying the oxidizing gas to the burner; a second heat exchanger transferring heat from the product of the reactor to the oxidizing gas at a first place on the first supply line that overlaps the second supply line; and a second dehydration unit

removing H.sub.2O gas from the product of the reactor at a second place disposed downstream of the first place on the first supply line.

7. The apparatus for producing glass according to claim 1, wherein the reactor has a metal catalyst promoting the reverse shift reaction.

8. An apparatus for producing glass that alternately repeats formation of a flame inside a glass melting furnace using a first burner and formation of a flame inside the glass melting furnace using a second burner, the apparatus for producing glass comprising: a first regenerator causing a reverse shift reaction of producing at least CO gas from CO.sub.2 gas and H.sub.2 gas to progress by recovering heat from a combustion gas emitted from the glass melting furnace while the second burner forms a flame inside the glass melting furnace and by discharging heat while the first burner forms a flame inside the glass melting furnace; the first burner forming a flame inside the glass melting furnace by burning a flammable gas containing the CO gas produced in the first regenerator and an oxidizing gas; a second regenerator causing a reverse shift reaction of producing at least CO gas from CO.sub.2 gas and H.sub.2 gas to progress by recovering heat from a combustion gas emitted from the glass melting furnace while the first burner forms a flame inside the glass melting furnace and by discharging heat while the second burner forms a flame inside the glass melting furnace; and the second burner forming a flame inside the glass melting furnace by burning a flammable gas containing the CO gas produced in the second regenerator and an oxidizing gas.

9. The apparatus for producing glass according to claim 8, wherein the oxidizing gas is pure oxygen gas or oxygen-enriched air.

10. The apparatus for producing glass according to claim 8, further comprising a circulation line supplying at least a part of the CO.sub.2 gas contained in the combustion gas emitted from the glass melting furnace from the second regenerator to the first regenerator while the first burner forms a flame inside the glass melting furnace.

11. The apparatus for producing glass according to claim 10, further comprising a dehydration unit, which is disposed in the middle of the circulation line, removing H.sub.2O gas from the combustion gas.

12. A method for producing glass comprising producing molten glass using the apparatus for producing glass according to claim 1.
